HDOC DAY – 51

Algorithm: Next Greater Element

1. Start.
2. Read the size of the array size.
3. Check whether size is positive.
4. Declare a variable-length integer array arr[size].
5. Read the elements of the array.
6. Pass the address of arr[] to the function next_greater().
7. In next_greater():
    - Repeat for each element arr[i].
    - Set next = -1.
    - Using an inner loop, start searching from index i + 1 toward the right.
    - Compare each right-side element with arr[i].
    - If arr[j] > arr[i]:
        - Store arr[j] in next.
        - Use break to stop searching because the nearest greater element has been found.
    - If no greater element is found, next remains -1.
    - Display next.
8. Separate the results using commas.
9. Stop.

C PROGRAM:

```
#include <stdio.h>

void next_greater(int *arr, int size)
{
   int i, j, next;

   for (i = 0; i < size; i++)
   {
      next = -1;

      // Search elements to the right
      for (j = i + 1; j < size; j++)
      {
         if (*(arr + j) > *(arr + i))
         {
            next = *(arr + j);
            break;   // Nearest greater element found
         }
      }

      printf("%d", next);

      if (i < size - 1)
      {
         printf(", ");
```

```
        }
    }

    printf("\n");
}

int main()
{
    int size, i;

    printf("Enter the size of the array: ");
    scanf("%d", &size);

    if (size <= 0)
    {
        printf("Invalid size.\n");
        return 0;
    }

    int arr[size];

    printf("Enter the array elements:\n");

    for (i = 0; i < size; i++)
    {
```

```
        scanf("%d", &arr[i]);
    }

    // Pass the address of the array
    next_greater(arr, size);

    return 0;
}
```

Test Case Table

Test	Size	Input Array	Expected Output
1	4	4 5 2 25	5, 25, 25, -1
2	4	13 7 6 12	-1, 12, 12, -1
3	4	1 3 2 4	3, 4, 4, -1
4	5	1 2 3 4 5	2, 3, 4, 5, -1
5	5	5 4 3 2 1	-1, -1, -1, -1, -1